

Pandas → 'load data' → dataframe  
↑ ↑  
↑ ↓ series

attribute/method } IT How to do filtering  
↑

df[ ]  
↑

describe()  
↓ max()  
↑ series → (col)

↓



groupby

→ join operation

n → groups

data

df.groupby('product').agg

Product

price

A

3

A

5

A

5

B

7

B

7

C

3

C

5

Price

A →

|   |   |
|---|---|
| A | 3 |
| A | 5 |
| A | 5 |

Sum → 13 Avg =  $\frac{13}{3}$   
max → 5

B →

|   |   |
|---|---|
| B | 7 |
| B | 7 |

14 + 7 →  $\frac{14}{2}$

C →

|   |   |
|---|---|
| C | 3 |
| C | 5 |

8 → 5 →  $\frac{8}{2}$

Q) find Avg price for each product

product

→

|     |  |
|-----|--|
| A → |  |
| B → |  |
| C → |  |



SQL

|     |     |
|-----|-----|
| C11 | C12 |
|     |     |

|     |     |
|-----|-----|
| C11 | C13 |
|     |     |

↓ join  
single table  
↑



|  |
|--|
|  |
|--|

left Right  
outer }

inner join  
↑



eg.

| <u>product-id</u> | Price(s.p) |
|-------------------|------------|
| 1                 | 40         |
| 2                 | 50         |
| 3                 | 40         |
| 4                 | 40         |
| 5                 | 50         |

df1

join

df2

| p.id | Product-catg |
|------|--------------|
| 1    | Electronics  |
| 1    | Electronics  |
| 2    | Grocery      |
| 3    | IT           |

left

look-up

df2.join(df1)

| P-Catg  | Price |
|---------|-------|
| Elect   | 40    |
| Elect   | 40    |
| Grocery | 50    |
| Elect   | NaN   |

join